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Paper ID

Mapping Rice Paddy Areas in Nueva Ecija Using Temporal Sentinel – 1 SAR Imagery

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Background of the Study

About 90% of the world's rice is produced in Asia, according to (Gutaker et al., 2020). As it is a staple crop in many regions, some countries produce rice for their supply and consumption, such as the Philippines. Out of 30 million hectares of agricultural land, about 4.81 million hectares are specified for rice production (Silva et al., 2017). In 2020, rice production in the Philippines reached 19.44 million metric tons, where the biggest contributions were the regions of Central Luzon, Cagayan Valley, and Ilocos (Agoot, 2020). The country's biggest rice producer is Region 3, Central Luzon. Due to its capability to produce millions of metric tons of rice, it is known as the Rice Capital of the Philippines and comprises seven provinces namely Aurora, Nueva Ecija, Bulacan, Pampanga, Tarlac, Bataan, and Zambales. Rice is cultivated in the Philippines primarily during the wet season, from June to early October mostly in naturally irrigated areas. Cultivation continues during the dry, to hot-dry season (late October to May) for areas with specific irrigation facilities (Raviz et al., 2016). Significantly, the researchers aim for the study to aid farmers in decision-making regarding water and moisture management, aiming to maximize harvests and minimize risks.

Objective of the Study

- Create a precise rice paddy map using supervised classification and utilize the use of cloud open-source platform for ease replication.
- Monitor and assess the change of rice paddy indices in municipality of Guimba on wet season planting.

Methodology

The study area of the rice mapping is Nueva Ecija Province, Philippines, located in the Central Luzon plains with an area of approximately 5,751.33 km² (Figure 1). The province experiences a humid subtropical climate, with mountainous terrain in the east, low hills in the north, and mainly plains in the central region. According to Global Forest Watch (2022), cropland is the main land use type in the study area, accounting for approximately 7.8%. And paddy rice is the main crop, accounting for 0.26%.

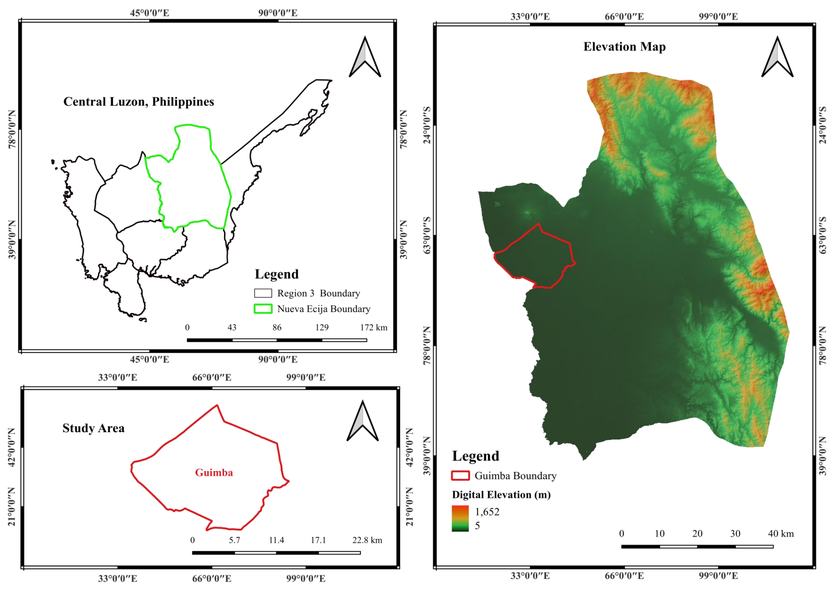

Figure 1. The study area.

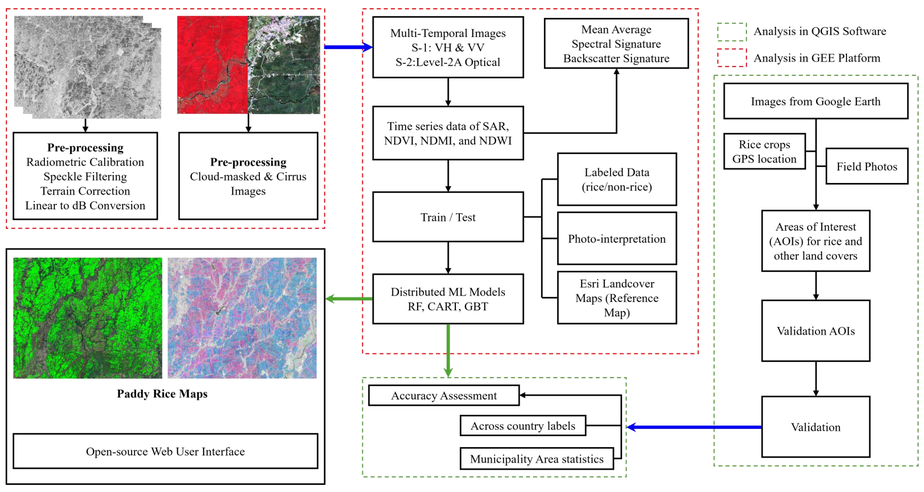

Figure 2. Process flow chart of this study.

Materials and Software

Results and Discussion

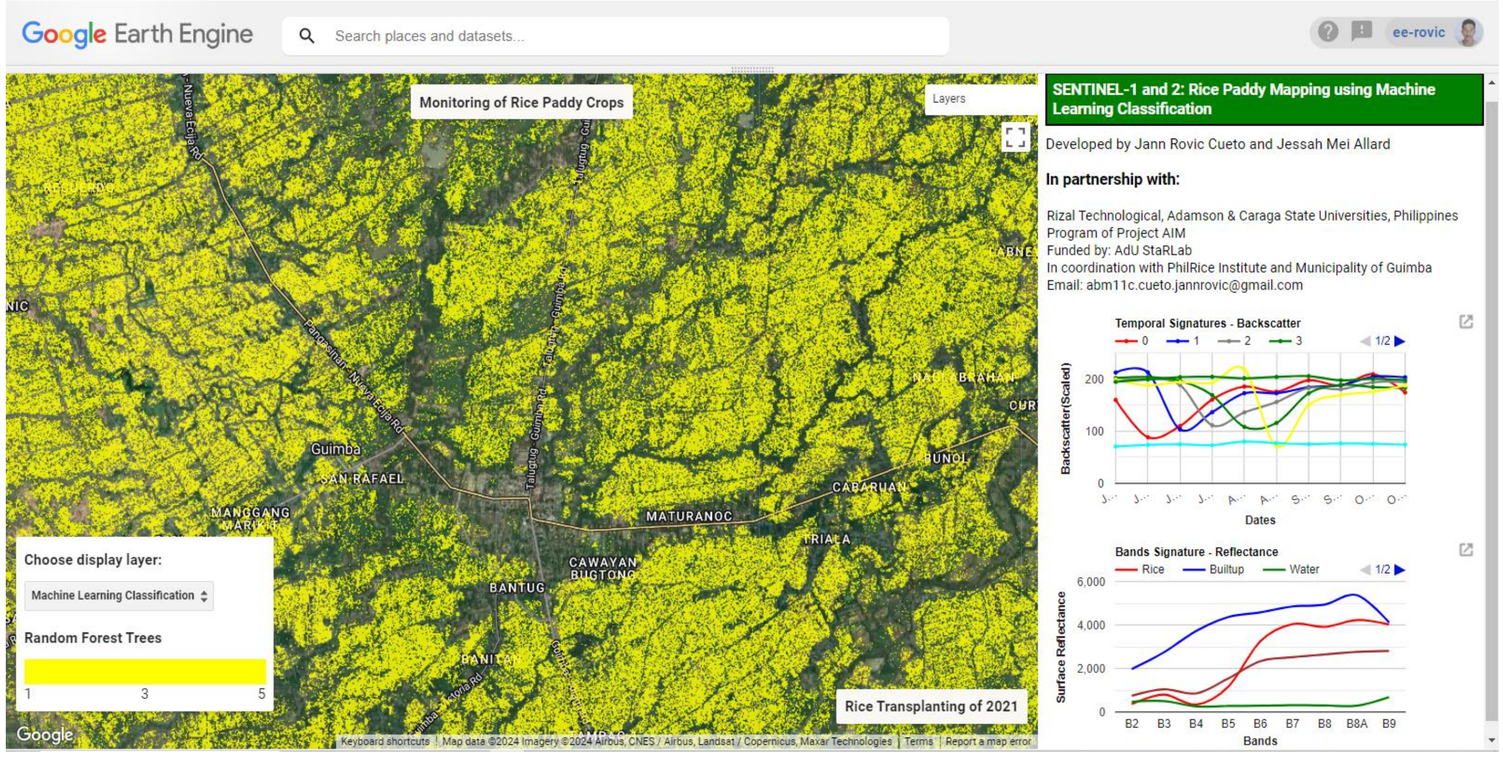

Figure 3. Open-source Web User Interface.

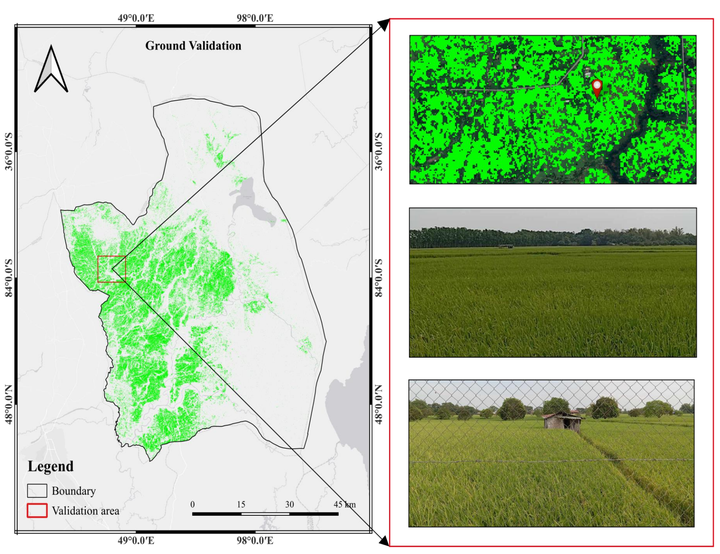

Figure 5 Ground Validation.

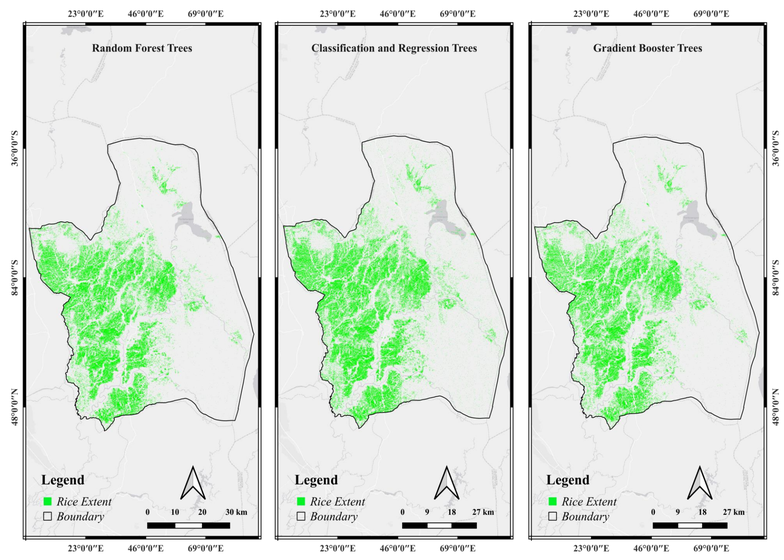

Figure 4. Rice paddy distribution classified by the models.

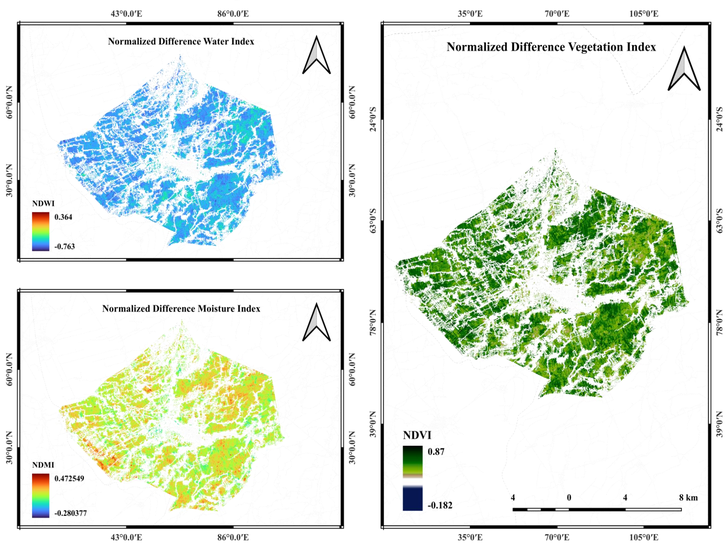

Figure 6. Rice paddy indices in Municipality of Guimba.

Class	Machine Learning Model					
	RF		CART		GBT	
	PA	UA	PA	UA	PA	UA
	%					
Rice	95	99.4	99	99.8	73.8	88.8
Non-Rice	66	66	87.7	99.3	66.2	84
Accuracy	97		99		98	
Kappa Coefficient	96.2		99		96	

Table 1. Accuracy assessment of the three models.

- Estimated rice paddy areas in each municipality per hectares.
- The random forest model classification result is more reliable than other model after the assessment of ground validation.
- Temporal indices monitoring in the classified rice paddy areas.
- Open-source web user interface and codes for ease replication.

Conclusion and Recommendation

The present study effectively utilized the Sentinel-1 images for the supervised rice classification and mapping in Guimba, Nueva Ecija. Upon the evaluation of results, the generated classification and map closely aligns with the actual coverage in the area which can be access on the web application platform that we created for ease visualization and replication of codes for analysis. While the results appeared to be promising, the researchers recommend improvements in classification accuracy by incorporating more annotated samples and exploring alternative machine learning models. Subsequently, the researchers integrated Sentinel-2 data indices to facilitate the monitoring of rice paddy health and growth throughout the rice cultivation period. Through the study's approach in utilizing remote sensing data for spatial and temporal monitoring, the results will be able to provide valuable insights for farmers in Guimba to optimize water management and enhance rice production.

References

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